DO NOW — RECALL

# Binary search: trace and code

Name: _____ Class: _____ Date: _____

**1. Does linear search require ordered input?**

**2. What must be true of the two input runs when merging them?**

**3. For seven elements, what is the last zero-based index?**

**4. Find the integer midpoint of low=2 and high=8.**

**DO NOW — CHECK / OBJECTIVES**

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

### 1. Model answer

No. It scans sequentially and works on unsorted arrays.

### 2. Model answer

They must already be sorted.

### 3. Model answer

6.

### 4. Model answer

5: 2 + (8-2)/2 = 5.

### Vocabulary — know these terms

precondition — Input assumption: the array is ordered.

low — Lower inclusive search index.

high — Upper inclusive search index.

mid — Index of the middle candidate.

O(log n) — Search repeatedly halves the candidate set.

NEW LEARNING

# Read, annotate and explain

### Sort before searching

Binary search relies on a sorted array; otherwise discarding half may lose the target.

### Inspect the middle

Compare target to the midpoint and keep only the half that could contain it.

### Stop correctly

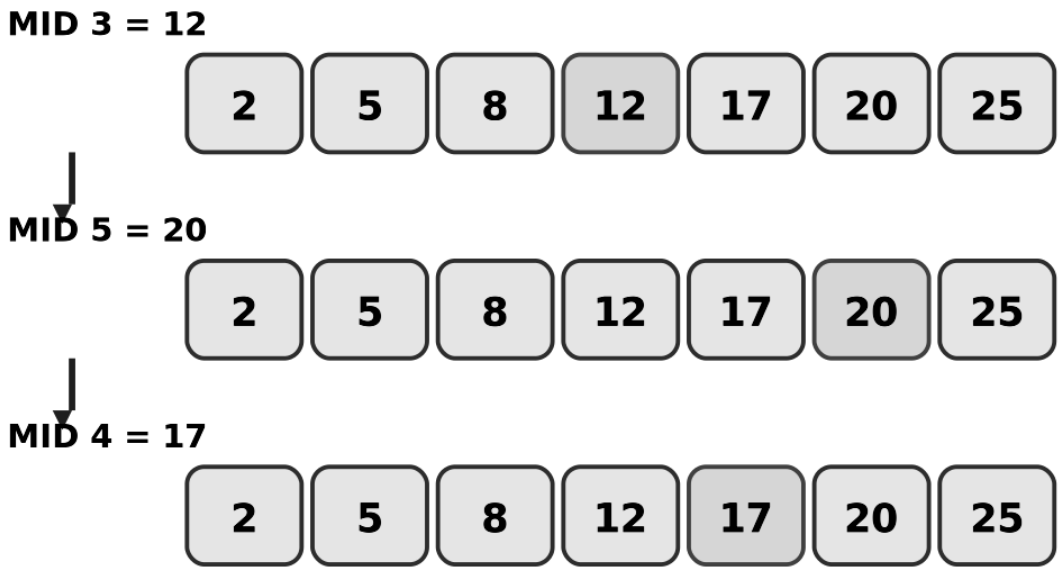
Return an index when found; return -1 if low becomes greater than high.

**Explain one step in your own words, using the correct technical terms.**

# New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**Read the code; predict its effect**




Worked algorithm states: MID 3 = 12 2,5,8,12,17,20,25; MID 5 = 20 2,5,8,12,17,20,25; MID 4 = 17 2,5,8,12,17,20,25

I DO — WORKED EXAMPLE

# Follow the worked method

**Search for 19 in [3,7,10,15,19,23,27]. Record every midpoint checked.**

### Predict first

Use inclusive low and high indexes.

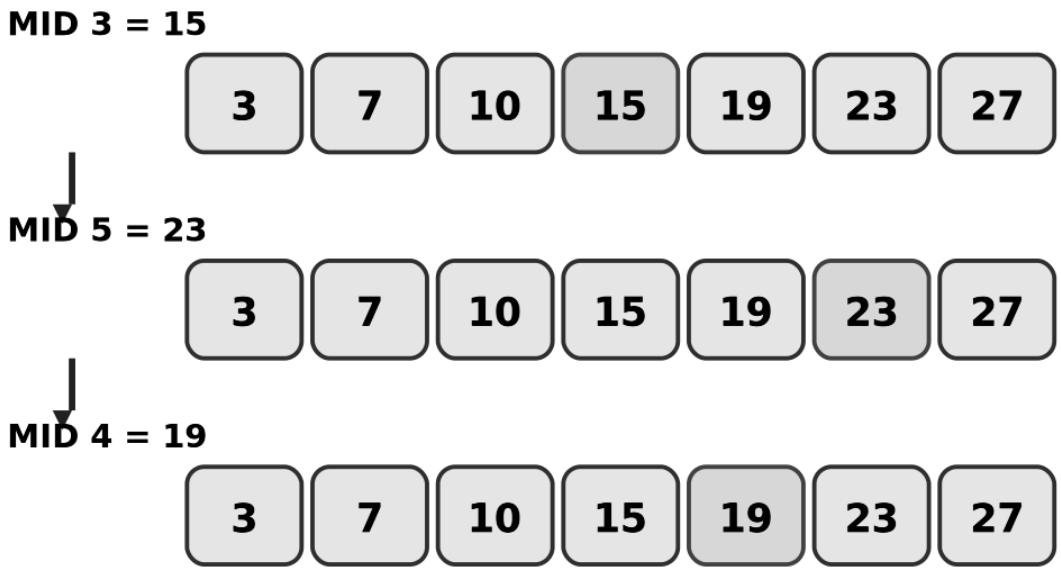
### Model solution

mid 3 → 15, so low=4; mid 5 → 23, so high=4; mid 4 → 19. Return 4.

# I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**Search for 19 in [3,7,10,15,19,23,27]. Record every midpoint checked.**




Worked algorithm states: MID 3 = 15 3,7,10,15,19,23,27; MID 5 = 23 3,7,10,15,19,23,27; MID 4 = 19 3,7,10,15,19,23,27

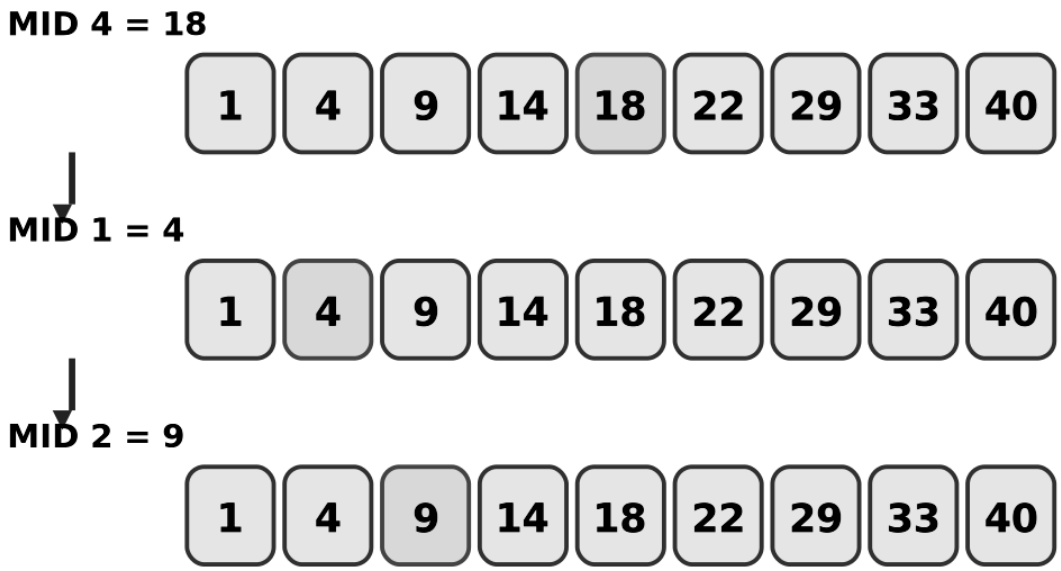
WE DO — GUIDED PRACTICE

# Try → Discuss → Improve

# We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**Search for 9 in [1,4,9,14,18,22,29,33,40].**




Worked algorithm states: MID 4 = 18 1,4,9,14,18,22,29,33,40; MID 1 = 4 1,4,9,14,18,22,29,33,40; MID 2 = 9 1,4,9,14,18,22,29,33,40

# WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

### Guided 1: Predict first Which midpoint is checked first?

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

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YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 1

Search for 8 in [2,5,8,11,14]. List midpoint indexes.

### Question 2

Search for 13 in [2,4,6,8,10,12,14]. List indexes checked and final return.

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 3

Explain why setting low = mid instead of mid + 1 can cause failure.

**EXIT TICKET / REFLECTION**

# Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

**Binary search is always safe on an unsorted array.**

**How confident are you? Circle one: Confident / Getting there / Need help**

Explain your choice, identifying one concept you can now use independently.

**GREEN PEN — Correct and explain**

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

# WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

### Guided 1 — original CHECK

Model solution

mid 4=18, then mid 1=4, then mid 2=9. Return index 2.

IMPROVE: Explain one correction to your own reasoning.